

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (CE)

December 2012

CE – 720 Object Orientation & Modeling Analytics (OO&MA)

Date: 22.12.2012, Saturday

Time: 010:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Rough work is to be done in the last page of main supplementary, please don't write anything on the question paper.
5. Indicate clearly, the option(s) you attempt along with its respective question no.
6. Figures to the right indicate marks.

SECTION-I

Q-1 Answer the following questions.

1. Why we need object oriented programming techniques? Give a brief 3
note on object relationships and association with an example.
2. Cannot we develop a system without models? Justify. How UML is 3
like a language? Explain the UML approach of software
development life cycle.
3. Discuss how structured approach is different from object oriented 2
approach based on the example of Library System.
4. What is an event? What is an edge triggered event? How a signal is 3
related to an event?

Q-2

- [A] Discuss the object oriented software development process in detail using 4
use case method. Describe the basic activities in Object Oriented Analysis
and explain how use case modeling is useful in analysis. Give the guideline
to identify actors and use cases in use-case diagram.
- [B] Compare and contrast between sequence and collaboration diagrams. 4
- [C] Discuss how an interaction diagram can be used to model the flow of 4
control by time ordering. Discuss how an interaction diagram can be used
to model the flow of control by organization.

OR

Q-2

- [A] How use cases are related to actors? Explain Use Case and Scenario. How 4
can we organize Use Case?
- [B] What is responsibility of a class? How classes can be used to model the 4
responsibility of a system? Give an example.
- [C] Explain Fork and Join with an example. Explain Swimlanes with an 4
example.

Q-3

- [A] Prepare the Data Flow Diagram for the Patient Information System as a part 4
of Hospital management System.
- [B] For a Social Networking Site, consider the functionality like :Stake holders 4
and Interests, Update User Account, Inform Friend about S.N. site, Send

Invite to other user, Process Friend Request, Search Friend in List, Send a message to a friend or in group, Update photo Album, Handle Login Failure. Identify the relevant use cases and prepare use-case diagram.

- [C] How to search the design pattern? How to document the design pattern? 4

OR

Q-3

- [A] Discuss how Activity Diagram can be used to model a workflow. 4

- [B] Explain the architecture of OOSE. 4

- [C] Justify and give your comments for the following Heuristics in brief: 4

1. Users of a class must be dependent on its public i/f, but a class should not be dependent on its users.
2. Minimize the no. of classes with which other classes collaborates.
3. Inheritance should be deep, deeper the better.
4. Factor the commonality of data behavior and/or interface as high as possible.

SECTION-II

Q-4

1. Define UML Diagrams. List out the different types of UML 3

Diagrams. 4

2. How to understand interfaces? What is a Package? What can be the owned elements of Packages? 4

3. Compare and contrast between sequence and collaboration diagram.

Q-5

- [A] What is the purpose of models in designing? Which diagram(s) show(s) the concept of Inheritance? What are the challenges in designing with Inheritance (single/multiple both)? 4

- [B] What are the different types of component? Differentiate between components and classes. 4

- [C] Define design pattern. Explain the need for patterns and explain about analysis pattern with its template. Discuss how design pattern solve design problems. 4

OR

Q-5

- [A] Under what situation activity diagram should not be used? Which diagrams are preferable under such situations? 5

Also give practical situations, when to use the following diagram:

- a. Use-Case Diagram
- b. Object Diagram
- c. Interaction diagram

- [B] Prepare object diagram showing at least 10 relationships among the following object classes. Include associations and qualified associations, aggregations, generalizations, and multiplicity. You may add additional objects. Also show attributes and operations. 4

School, playground, principal, schoolboard, classroom,

book, student, teacher, canteen, restroom, computer, desk, chair.

- [C] What the use is of "include" and "extends" relationships in use-case diagram? Draw the use-case diagram for Online Admission Process for Engineering Students in Gujarat. 3

Q-6

- [A] Discuss how activity diagram can be used to model a workflow. Differentiate between action state and activity state. 4

- [B] How a state chart diagram can be used to model a reactive object? Compare and contrast between activity and state chart diagram. 4

OR

- [B] What is a state diagram? Explain the characteristics of the system that can be identified by studying the state diagram of that System. 4

- [C] Define collaboration and sequence diagrams. Draw the UML interaction diagrams for Hotel Management System. There are two types of customers: Tour-group customers and Individual customers. Both can book, cancel, check-in and check-out of a room by Phone or via the Internet. There are booking process clerk and reception staff who manages it. A customer can pay his bill by credit card or pay utility bill. 4

OR

- [C] Draw the class diagram and use case diagram for Railway Reservation System. Customer can reserve, cancel, adjust and do inquiry of journey tickets. Travel agent can do bulk reservation, cancellations, adjust and do inquiry of journey tickets and needs to be notified of important events like introduction of new or seasonal trains. You may add to these functionalities. 4